Feasibility Analysis and System proposal

UNIVERSITY OF COLOMBO SCHOOL OF COMPUTING

Learning outcomes

- Identify feasibility checkpoints in the system's life cycle
- Identify alternative system solutions
- Define and describe six types of feasibility and their respective criteria
- Perform various cost-benefit analysis
- Write suitable system proposal reports for different audiences
- Plan for a formal presentation to system owners and users

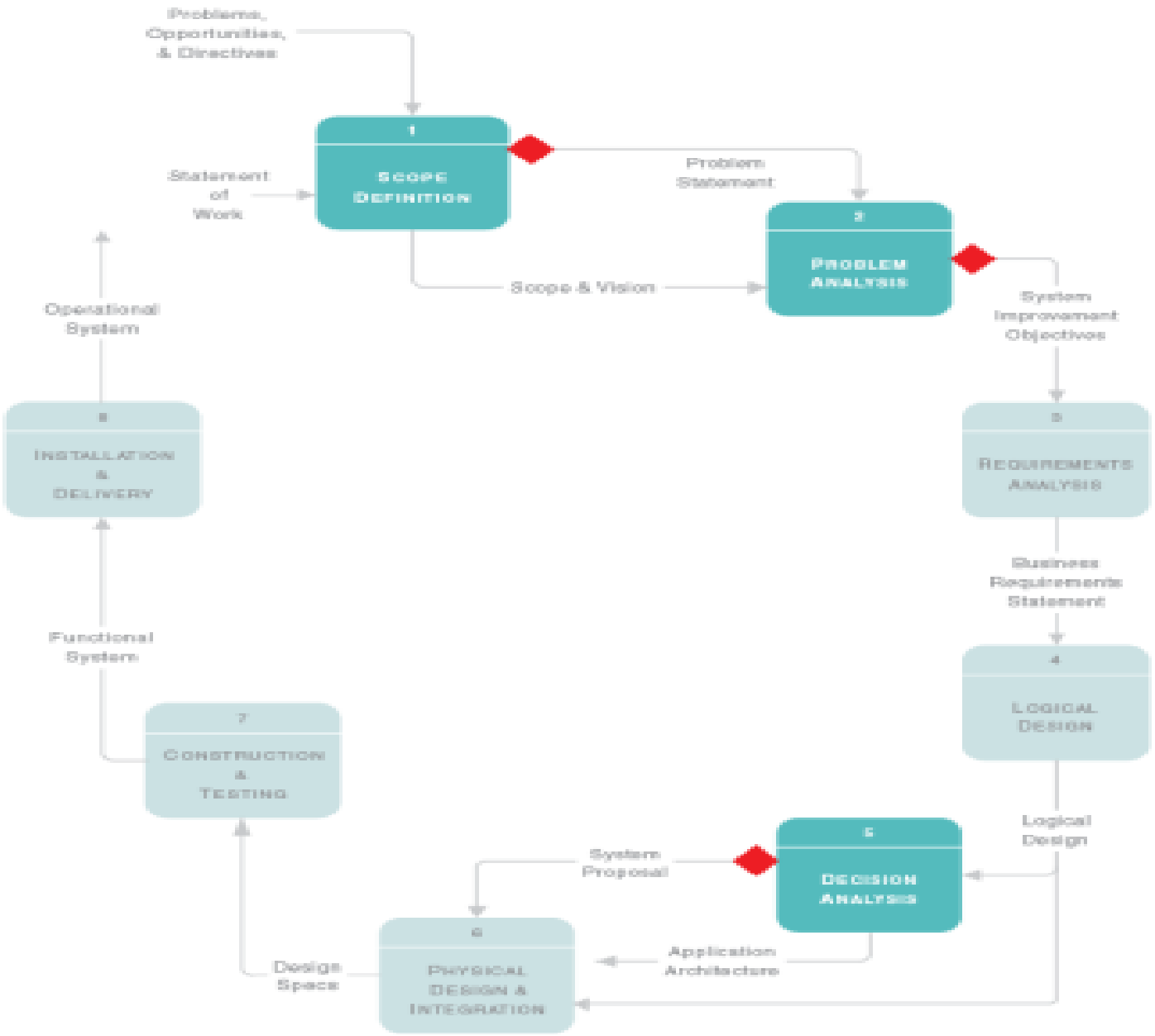
Feasibility Study

- **Feasibility**
  - The measure of how beneficial / practical an information system will be to an organization
  - Should be measured through out the life-cycle

- **Feasibility Analysis**
  - The process by which the feasibility is measured
  - An ongoing evaluation of feasibility at various checkpoints in the life cycle

Feasibility Checkpoints in the Software Development Life Cycle

- Feasibility of a project can be changed during the system development.
- To reevaluate feasibility, there are different checkpoints in the development.
- A project may be **canceled**, **revised** or **continued** at any checkpoint, despite whatever resources have been spent.


Problems, Opportunities, & Directives
1
Scope Definition
Statement of Work
Problem Statement
2
Problem Analysis
Scope & Vision
System Improvement Objectives
3
Requirements Analysis
Business Requirements Statement
4
Logical Design
Logical Design
5
Decision Analysis
System Proposal
6
Physical Design & Integration
Application Architecture
Design Space
7
Construction & Testing
Functional System
8
Installation & Delivery
Operational System

Typical Feasibility Checkpoints

- Systems Analysis – Scope Definition Checkpoint
- Systems Analysis – Problem Analysis Checkpoint
- Systems Design – Decision Analysis Checkpoint

Scope Definition Checkpoint

- Measure of the urgency of the problem.
- Find the first-cut estimate of development costs.
- Answer the question,
  - Do the problems warrant the cost of a detailed study & analysis of the current system?

Problem Analysis Checkpoint

- Occurs after a more detailed study and problem analysis of the current system
- Can make a better estimate of the development cost and benefits

Decision Analysis Checkpoint

- Represent major feasibility analysis activities
- Charts one of many possible implementations as the target
- Alternate solutions are defined in terms of,
  - Input/Output methods
  - Data storage methods
  - Hardware requirements
  - Software requirements
  - Processing methods
  - People implications

Decision Analysis Checkpoint

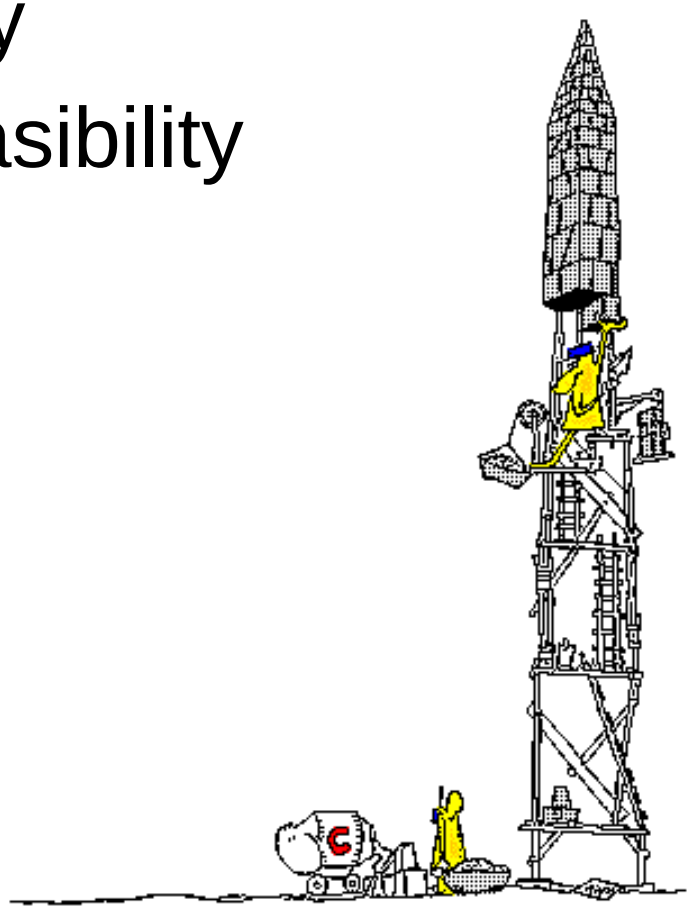
Range of options that can be evaluated by the analyst:

- Do nothing – leave the current system- baseline option
- Re-engineer the manual process -
- Enhance existing computer processes
- Purchase packaged software
- Design and construct a new computer-based system (Standalone)
- Design and construct a new computer-based system (Web-based) etc.

After defining these options, each option should be analyzed.

Six Tests for Feasibility

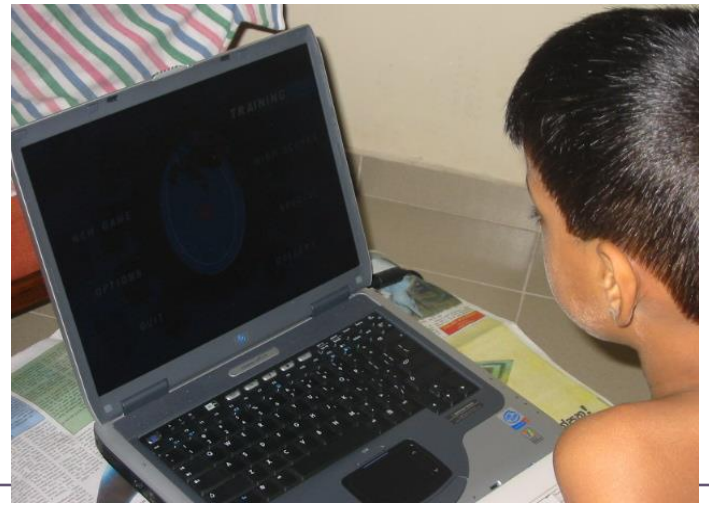
- Operational Feasibility
- Cultural / Political Feasibility
- Technical Feasibility
- Schedule Feasibility
- Economic Feasibility
- Legal Feasibility



Operational feasibility.

- A measure of how well a solution meets the identified system requirements to solve the problem.
- Take advantage of the opportunities identified during the scope definition and problem analysis phases.
- If given what is now known about the problem and the cost of the solution ,

  the problem is still worth solving?



Cultural (or Political) Feasibility

- A measure of how well the solution will be accepted in an organizational climate
- Deals with how the end users feel about the proposed system.
- Evaluates whether a system will work in an organizational climate.
- Questions:
  - Does management support the system?
  - How end users feel about the role in the new system?
  - Will end users/managers not use the system? Reason? Etc.

Technical feasibility.

- A measure of the
  - Practicality of a technical solution
  - Availability of technical recourses and expertise

- Addresses 3 major issues:
  1. Is the proposed technology or solution practical? – mature enough?
  2. Do we currently possess the necessary technology
  3. Do we possess the necessary technical expertise?

Schedule feasibility

- A measure of how reasonable a project time table is.
  - Can the solution be designed and implemented within an acceptable time period?
  - how much time is available to build the new system?
  - when it can be built ?

- Mandatory / Desirable deadlines.

Legal Feasibility

- A measure of how well a solution can be implemented within existing legal and contractual obligations.

- understand potential legal and contractual ramifications of the system

  * copyright law

  * code ownership (if developed with outside assistance) -- be VERY specific

  * labour laws

  * foreign trade, and labour regulations

  * Financial & Accounting standards

  * governmental constraints, and pending legislation

Economic feasibility.

- a measure of the cost-effectiveness of a project
  - Is the solution cost-effective?
  - Whether the solution will pay for itself?
  - How profitable the solution is?

- Once the specific requirements and solutions have been identified
  - Weight the costs and benefits of each alternative (Cost benefit Analysis)

e.g. Personnel cost, Computer cost, Training, Software, Tangible and Intangible benefits

Cost Benefit Analysis

- Determines the cost effectiveness of a project or solution
- The purpose of a cost/benefit analysis is to answer questions such as:

  - Is the project justified (because benefits outweigh costs)?
  - Can the project be done, within given cost constraints?
  - What is the minimal cost to attain a certain system?
  - What is the preferred alternative, among candidate solutions?

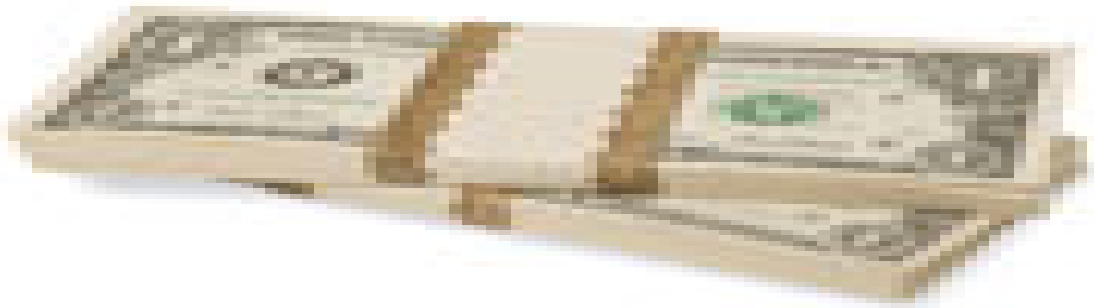
How much will the system cost?

- Two types of costs,

  1. Developing the system
     - Can be estimated from the outset of a project
     - Should be refined at the end of each phase
     - One time costs (will not recur after the project has been completed
  2. Operating a system
     - Can be estimated only after specific computer-based solutions have been defined
     - Recur throughout the lifetime of the system

How much will the system cost?

- System development Cost Categories
  - Personnel costs
  - Computer Usage
  - Training
  - Supply, duplication, and equipment costs
  - Cost of any new computer equipment and software



What benefits will the system provide?

- Benefits
  - Increase profit
  - Decrease costs
  - Can be classified as
    - Tangible benefits – a benefit that can be easily quantified.
    - Intangible benefits – a benefit that is believed to be difficult or impossible to quantify

Feasibility Analysis of Candidate systems

- During the decision analysis phase of system analysis,
  - Identifies candidate system solutions
  - Analyses the solution for feasibility
- Can use two methods to compare and contrast candidate system solutions
  - Candidate System Matrix
  - Feasibility Analysis Matrix

} **Use A Matrix Format**

Candidate Systems Matrix

- Used to document similarities and differences between candidate systems
  - Compare candidate systems
  - Offers no analysis
  - Columns represent candidate solutions
  - Rows represent characteristics that differentiate the candidates

Candidate Systems Matrix

- Example

	Candidate 1 Name	Candidate 2 Name	Candidate 3 Name
Stakeholders			
Knowledge			
Processes			
Communications			

Template

Candidate Systems Matrix

- Example

	Candidate 1 Name	Candidate 2 Name	Candidate 3 Name
Stakeholders			
Knowledge			
Processes			
Communications			

Identify how the system will interact with people, and other systems

Template

Candidate Systems Matrix

- Example

	Candidate 1 Name	Candidate 2 Name	Candidate 3 Name
Stakeholders			
Knowledge			
Processes			
Communicatio			

Identify how data stores be implemented,
inputs will be captured,
outputs will be generated

Template

Candidate Systems Matrix

- Example

	Candidate 1 Name	Candidate 2 Name	Candidate 3 Name
Stakeholders			
Knowledge			
Processes			
Communications			

Identify How (manual) business process will be modified, how computer processes will be implemented

Template

Candidate Systems Matrix

- Example

	Candidate 1 Name	Candidate 2 Name	Candidate 3 Name
Stakeholders			
Knowledge			
Processes			
Communications			

Identify how processes and data will be distributed.

Template

Feasibility Analysis Matrix

- Used to rank candidate systems
  - Columns represent candidate response
  - Rows correspond to the feasibility criteria
  - Cell contain the feasibility assessment notes for each candidate

	Weighting	Candidate1	Candidate2	Candidate3
Description				
Operational Feasibility				
Cultural Feasibility				
Technical Feasibility				
Economic Feasibility				
Schedule Feasibility				
Legal Feasibility				
Weighted Score				

Eg. : - Refer Ref 3 page 429-

The System Proposal

- A report / presentation of a recommended solution
- Usually a formal written report or oral presentation
- Intended for system owners and users

Formal written report

Oral presentation

System owners
And
End Users

Formal Presentation

- A special meeting
- Used to sell new ideas and gain approval for new systems
- Can be also used for
  Sell a new system, sell new ideas, sell change, head off criticisms, address concerns, verify conclusion, clarify facts and report progress.